

Notes: Gurun and Butler (JF, 2012)

Don't Believe the Hype: Local Media Slant, Local Advertising, and Firm Value

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1 Big picture

- **Question.** Do local newspapers report on nearby firms in a systematically more favorable way, what explains that local media slant, and does it affect firm value?
- **Setting.** The paper studies **U.S. newspaper coverage from 2002–2006**, using stories from **eight large local newspapers**, the **Wall Street Journal**, and **Dow Jones Newswire**, matched to firm-level advertising, accounting, analyst, ownership, and return data.
- **Core challenge.** A local paper may look more positive for many reasons that are not necessarily bias: local readers may prefer less negative stories, local journalists may face reporting constraints, or firms that advertise more may receive softer treatment. The paper tries to distinguish among these channels.
- **Main contribution.** The paper builds a tractable **text-based measure of media slant**, links it to **localness** and **advertising expenditures**, and then studies whether **abnormal local slant** is associated with stock returns and market value.
- **Main hypotheses.** The paper considers three candidate explanations for positive local slant:
 1. **Catering hypothesis:** local papers cater to local readers, especially employees of local firms.
 2. **Constrained reporting hypothesis:** local papers have fewer resources and therefore reproduce management-friendly information.
 3. **Advertising hypothesis:** local papers soften coverage of firms that spend more on local advertising.
- **Why it matters.** Finance research often treats media coverage as information. This paper argues that the *content* of media stories can itself be endogenous to conflicts of interest. That matters for both **corporate governance** and **asset pricing**.
- **Main empirical findings.**
 - When local media report on local firms, they use **fewer negative words** than when they report on nonlocal firms.
 - In baseline regressions, the **Local Firm** coefficient is positive and statistically significant; with firm-month fixed effects it is about **0.515**, with the richer baseline controls it is about **0.192**, and with firm fixed effects it is about **0.325**.
 - The evidence does **not** support the employee-catering story and does **not** provide clean support for the constrained-reporting story.
 - **Local advertising** is strongly related to more positive local slant. In IV estimates, the local advertising effect becomes much larger, while the national advertising effect becomes negligible.
 - A propensity-score design and a Craigslist quasi-experiment point in the same direction: **local advertising causes local slant**.
 - A long-short portfolio based on abnormal local slant earns about **5.52% per year** after standard factor controls.
 - In firm-value regressions, a one-standard-deviation increase in annual local slant is associated with about **4.29% higher firm value**.

- The valuation effect is strongest for firms that are **small, held more by individuals, opaque, or hard to arbitrage**.
- **Economic takeaway.** Local media do not just passively transmit information. They can act as **cheerleaders** for nearby firms, especially when local advertising dollars are at stake, and this positive slant appears to affect prices most where local investors are likely to matter most.

2 Data and measurement (key objects)

2.1 News sample and media outlets

- **Sources.** The paper uses **Factiva** to collect lead paragraphs of firm-specific news stories from:
 - **local newspapers:** Boston Globe, Chicago Sun-Times, Denver Post, Pittsburgh Post-Gazette, San Francisco Chronicle, Seattle Post-Intelligencer, St. Louis Post-Dispatch, and Washington Post;
 - **national outlets:** Wall Street Journal and Dow Jones Newswire.
- **Sample period. 2002–2006.**
- **Raw scale.** The authors retrieve more than **330,000** qualifying stories containing more than **100 million words**.
- **News inclusion rule.** A story must:
 - contain the firm’s official name in the headline and within the first 25 words of the lead paragraph, and
 - contain at least **50 words**.
- **Composite story construction.** All qualifying stories for a given **firm–month–media outlet** are combined into a **composite story**. This is the main unit of observation in the slant regressions.

2.2 Advertising, firm, and market data

- **Advertising data.** Monthly ad spending comes from **TNS Media Intelligence (TNSMI)**.
- **Advertising channels used in the paper.**
 - **National newspaper advertising:** New York Times, USA Today, and Wall Street Journal.
 - **Local/newspaper advertising:** advertising in more than **250** daily and Sunday newspaper editions and Sunday magazines.
- **Firm and finance data.**
 - CRSP/COMPUSTAT for accounting and returns,
 - First Call for analyst following,
 - SEC 13-F filings for institutional ownership.

- **Advertising matching.** TNSMI reports spending by *brand*; the authors aggregate to the firm level and match firms conservatively to CRSP identifiers.
- **Matched advertising sample.** Out of 9,604 CRSP company names in the period, the matching procedure yields **1,457 unique company names** in TNSMI.

2.3 Location, localness, and market definitions

- **Firm location.** Headquarters zip codes come from **Bloomberg**.
- **Newspaper location.** Newspaper headquarters are collected from newspaper websites.
- **Distance measure.** Distances between firms and newspapers are computed from latitude/longitude information using Census Gazetteer data.
- **Local Firm definition.** A firm is local to a newspaper if its headquarters is within **100 miles** of the newspaper. A **250-mile** radius gives similar results.

2.4 How slant is measured

- **Dictionary.** The paper uses the **Loughran–McDonald** financial dictionary.
- **Word counts.**
 - **2,337** words are classified as negative.
 - **353** words are classified as positive.
- **Preferred slant measure.** The main measure uses the **fraction of negative words** in a composite story, multiplied by -100 . Higher values therefore mean **more positive slant** (that is, fewer negative words).
- **Why negative words only?** The authors argue that negative information likely has stronger effects than positive information, so the negative-word share is the more informative object.
- **Range.** Slant is bounded between -100 and **0**, with values closer to zero meaning more favorable coverage.

2.5 Main sample characteristics and empirical objects

- **Firm sample size.** The main empirical sample contains **5,330 CRSP-listed firms**, though some tests use smaller samples because advertising data are not always available.
- **Average coverage.** A company appears in the news sample about **2.01 times per year** on average.
- **Firm size.** Mean assets are about **\$3.6 billion**; the median is about **\$379 million**.
- **Information environment.** Average analyst following is about **3.65**, and average institutional ownership is about **47%**.
- **Advertising frequency.** Roughly **42%** of firms have nonzero advertising spending in a typical year.

- **Average newspaper advertising.**
 - local newspaper advertising: about **\$0.29 million** annually,
 - national newspaper advertising: about **\$1.49 million** annually.

2.6 Key descriptive facts

- **Wall Street Journal tone.** Among the outlets in the sample, Wall Street Journal stories are the most negative on average, with mean slant around -3.19 .
- **Local versus nonlocal in raw means.** In raw outlet-by-outlet comparisons, local and nonlocal stories are not always dramatically different, which is why the paper emphasizes multivariate regressions.
- **Interpretation.** The key empirical object is not simply whether local stories look nicer on average, but whether **the same firm’s coverage becomes more favorable when the outlet is local and when local advertising incentives are stronger.**

3 Models and empirical specifications (all equations + interpretation)

3.1 Story-level slant measure

The core textual measure is

$$Slant_{itm} = -100 \times \frac{\# \text{ negative words in composite story}_{itm}}{\# \text{ total words in composite story}_{itm}},$$

where:

- i indexes the firm,
- t indexes the month,
- m indexes the media outlet.
- **Interpretation.** Because the ratio is multiplied by -100 , a *higher* value means **less negative and therefore more favorable** language.
- **Economic use.** This is the paper’s basic measure of “hype.”

3.2 Baseline localness regression

A clean specification compares the same firm’s news across outlets in the same month:

$$Slant_{itm} = \alpha + \beta LocalFirm_{im} + \mu_m + \phi_{it} + \varepsilon_{itm},$$

where:

- $LocalFirm_{im} = 1$ if firm i is headquartered within 100 miles of outlet m ,

- μ_m are media fixed effects,
- ϕ_{it} are firm-month fixed effects.
- **Interpretation.** This is the paper’s cleanest comparison. It asks: holding the firm and month fixed, does a **local** outlet write less negatively than a **nonlocal** outlet?
- **Main result.** The answer is yes: the coefficient on *LocalFirm* is strongly positive.

3.3 Broader slant regression with controls

The richer baseline is

$$Slant_{itm} = \alpha + \beta LocalFirm_{im} + X'_{it}\Gamma + \mu_m + \lambda_{ind(i)} + \tau_t + \varepsilon_{itm},$$

where X_{it} includes:

- log market value,
- ROA,
- book leverage,
- analyst following,
- institutional ownership,
- momentum,
- idiosyncratic volatility,
- current return,
- sales growth,
- log employees,
- employees divided by local population,
- number of segments,
- *Plain English* readability.
- **Interpretation.** This regression tests whether local slant survives after controlling for firm characteristics, firm information environment, and newspaper-specific style.
- **How the paper uses it.** The employee variables are used to test the **catering hypothesis**; the number of segments and readability variables are used to test the **constrained reporting hypothesis**.

3.4 Advertising-hypothesis regression

To test whether ad spending influences tone, the paper estimates

$$Slant_{itm} = \alpha + \beta LocalFirm_{im} + \delta \log(LocalAdvertising_{it}) + \nu \log(NationalAdvertising_{it}) + X'_{it}\Gamma + FE + \varepsilon_{itm}.$$

- **Interpretation.** A positive δ means that firms spending more in local newspapers receive more favorable local media language.
- **Why both ad variables enter.** This separates local newspaper incentives from national newspaper advertising demand.
- **OLS result.** Both local and national advertising are positively related to slant in OLS, but this may reflect endogeneity.

3.5 Instrumental-variables specification

Because advertising is endogenous, the paper instruments for firm-level ad spending using industry-year averages:

$$\begin{aligned} \log(\text{LocalAdvertising}_{it}) = & \pi_0 + \pi_1 \log(\text{IndustryLocalAdvertising}_{-i,st}) \\ & + \pi_2 \log(\text{IndustryNationalAdvertising}_{-i,st}) + X'_{it}\Pi + FE + u_{it}, \end{aligned}$$

$$\begin{aligned} \log(\text{NationalAdvertising}_{it}) = & \rho_0 + \rho_1 \log(\text{IndustryLocalAdvertising}_{-i,st}) \\ & + \rho_2 \log(\text{IndustryNationalAdvertising}_{-i,st}) + X'_{it}\Gamma^N + FE + v_{it}, \end{aligned}$$

and then uses the fitted values in the second-stage slant regression.

- **Instrument.** Industry-year averages are constructed at the **two-digit SIC** level and exclude the firm's own contribution.
- **Interpretation.** The IV isolates the part of ad spending driven by broader industry advertising patterns rather than the firm's own news quality.
- **Main result.** The **local advertising** coefficient becomes much larger and remains significant, while the **national advertising** coefficient becomes insignificant.

3.6 Propensity score matching design

The matching design estimates the probability that a firm advertises in local newspapers:

$$\Pr(\text{LocalAdvertising}_{it} > 0) = \Phi(Z'_{it}\theta),$$

where Z_{it} contains the same baseline controls and fixed effects used in the main slant regressions.

- **Interpretation.** Firms with similar observables are matched, and the paper compares slant between those that do and do not advertise locally.
- **Treatment.** Positive local advertising expenditure.
- **Estimand.** Average treatment effect on slant.

3.7 Craigslist quasi-experiment

To study changes in newspaper dependence on advertising revenue, the paper estimates

$$\begin{aligned} Slant_{itm} = & \alpha + \beta \text{Craigslist}_t + \delta \log(\text{LocalAdvertising}_{it}) \\ & + \nu \log(\text{NationalAdvertising}_{it}) \\ & + \theta [\log(\text{LocalAdvertising}_{it}) \times \text{Craigslist}_t] + \psi [\log(\text{NationalAdvertising}_{it}) \times \text{Craigslist}_t] \\ & + X'_{it}\Gamma + FE + \varepsilon_{itm}, \end{aligned}$$

where $\text{Craigslist}_t = 1$ after Craigslist enters Pittsburgh and St. Louis in October 2003.

- **Interpretation.** If Craigslist weakens newspapers' classified-ad revenue, local newspapers may become more dependent on corporate advertisers.
- **Testable implication.** The coefficient θ should be positive if local-advertising sensitivity increases after Craigslist entry.

3.8 Return-based pricing test

The paper defines **abnormal local media slant** as

$$\text{AbnormalLocalSlant}_{it} = \text{LocalSlant}_{it} - \text{NationalSlant}_{it},$$

and then forms a portfolio that is:

- long stocks with **low** abnormal local slant,
- short stocks with **high** abnormal local slant.

The resulting long-short return is evaluated using factor regressions of the form

$$R_t^{LS} = \alpha + b_M(RM - RF)_t + b_S SMB_t + b_H HML_t + b_U UMD_t + b_L LIQ_t + b_N NOCOV_t + \varepsilon_t.$$

- **Interpretation.** If overly positive local hype inflates prices, then high-abnormal-local-slant stocks should subsequently underperform.
- **NOCOV factor.** This controls for the known pricing effect of **media coverage itself**, separate from tone.

3.9 Firm-value regression

The main valuation regression is

$$\begin{aligned} \log Q_{it} = & \alpha + \beta_L \text{AnnualLocalSlant}_{it} + \beta_N \text{AnnualNationalSlant}_{it} \\ & + \gamma_L \text{AnnualLocalCoverage}_{it} + \gamma_N \text{AnnualNationalCoverage}_{it} \\ & + X'_{it}\Gamma + \lambda_{ind(i)} + \tau_t + \varepsilon_{it}. \end{aligned}$$

Here:

- Q_{it} is Tobin's Q,

- annual slant is the average of monthly slant measures,
- annual coverage is the number of stories in a year,
- X_{it} includes size, ROA, CAPX-to-sales, book leverage, R&D-to-sales, advertising-to-sales, analyst following, institutional ownership, momentum, idiosyncratic volatility, and sales growth.
- **Interpretation.** This asks whether firms with more favorable local-language coverage have higher valuations, *holding fixed* national slant and the quantity of local and national media coverage.
- **Why national slant matters.** The authors treat national slant as a control for more general news/fundamentals, so the local-slant coefficient is meant to capture the incremental role of local hype.

4 Identification and instruments (what makes the estimates credible)

4.1 What a naive comparison would miss

- Raw average differences between local and nonlocal stories are not enough because outlets differ in writing style, firms differ in visibility, and news events differ in timing.
- Advertising is clearly endogenous: firms with more visible brands or better news may both advertise more and receive better coverage.
- Valuation tests are also difficult because both **tone** and **coverage** may matter for firm value.

4.2 What identifies the local-media slant effect

- The firm-month fixed-effect regression identifies the localness effect from **within-firm, within-month, across-outlet** variation.
- Media fixed effects absorb outlet-specific editorial style.
- Firm fixed effects in alternative specifications absorb time-invariant firm-specific omitted variables.
- Robustness to broader distance cutoffs helps show the result is not driven by one arbitrary geographic definition.

4.3 Why the catering and constrained-reporting tests matter

- If the effect came mainly from employee demand, then firms with more local employees should receive more favorable local coverage.
- If the effect came from constrained journalism, then more complex firms should generate more favorable stories because reporters would lean more on management language.
- The paper finds little support for either channel, which sharpens the case for the advertising mechanism.

4.4 Instrumental-variables logic

- The instruments are **industry-year average local and national advertising expenditures**, excluding the firm's own contribution.
- Relevance comes from the idea that a firm's advertising choice co-moves with its industry's broader advertising patterns.
- Exclusion is argued from the idea that industry-wide advertising in local or national papers should not directly change the tone of one particular paper's coverage of one particular firm, except through the firm's own advertising.
- First-stage strength is strong in the paper's reported statistics: F-statistics are about **41** for local advertising and **158** for national advertising.

4.5 Alternative causal checks

- **Propensity score matching** reduces concerns that observable firm characteristics are driving both advertising and favorable coverage.
- **Craigslist entry** provides a quasi-natural experiment: when newspapers lose classified-ad revenue, they become more sensitive to corporate advertising, and the slant-for-local-advertising relation strengthens.
- The fact that both alternative approaches point in the same direction is one of the strongest parts of the paper.

4.6 What the valuation evidence does and does not identify

- The asset-pricing and Tobin's Q results show that local slant is **priced or associated with valuation**, especially for firms where local investors are more likely to be marginal investors.
- But these tests are not as cleanly causal as the advertising-slant tests. They are best read as evidence that local slant is **economically meaningful**, not as a fully structural estimate of sentiment-driven mispricing.
- The paper's strategy is to control for national slant and both local and national coverage, thereby focusing on the incremental relation between **local tone** and value.

5 Tables

5.1 Table I: descriptive statistics on companies

- **Read.** The sample consists of relatively large, visible public firms compared with the full COMPUSTAT universe.
- Mean assets are about **\$3.6 billion**; median assets are about **\$379 million**.
- Average analyst following is about **3.65**, and average institutional ownership is about **47%**.

- Average local newspaper advertising is about **\$0.29 million** per year; average national newspaper advertising is about **\$1.49 million**.
- **Takeaway.** The paper is not studying obscure microcaps only; it studies reasonably visible firms, which makes the slant result harder to dismiss as a niche phenomenon.

Table I
Descriptive Statistics on Companies

This table summarizes the characteristics of the firms analyzed in the paper. The sample period is 2002 to 2006. The unit of observation is the firm-year. For the COMPUSTAT column, we code missing observations of *Advertising Expenditures*, *R&D Expenses*, *CAPX*, *Analyst Following*, *Institutional Ownership*, *Annual National*, and *Local Media Coverage* as zero. A description of the variables is provided in Appendix C.

	Mean	Median	SD	75%ile	25%ile	COMPUSTAT Mean
Market value of equity (million \$)	3,567	379	15,863	1,509	90	3,126
Book leverage	0.429	0.463	0.750	0.672	0.234	0.320
Momentum	0.220	0.102	0.826	0.381	-0.153	0.210
Idiosyncratic risk	0.131	0.103	0.098	0.163	0.068	0.121
R&D expenses/sales	0.045	0.000	1.800	0.058	0.000	0.042
Advertising expenditures/sales	0.011	0.000	0.039	0.007	0.000	0.009
CAPX/sales	0.467	0.026	42.585	0.057	0.010	0.055
<i>Tobin's Q</i>	1.517	1.102	1.744	1.818	0.656	1.618
ROA	0.008	0.024	0.320	0.077	-0.018	0.017
Analyst following	3.645	1.000	5.254	5.000	0.000	1.930
Institutional ownership (%)	0.479	0.509	0.321	0.761	0.171	0.351
Number of employees	10,092	1,118	41,340	5,400	0.256	9,019
Employees/population (×1,000)	0.356	0.004	3.851	0.025	0.001	0.293
Number of segments	1.713	1.000	1.302	2.000	1.000	1.457
Plain English	0.668	0.474	2.219	2.193	-0.908	0.671
Annual nat'l and local media coverage	2.014	1.609	1.752	2.565	1.099	3.996
Local newspaper advertising (million \$)	0.286	0.000	3.214	0.000	0.000	n/a
Nat'l newspaper advertising (million \$)	1.493	0.000	19.139	0.000	0.000	n/a
Slant	-1.689	-1.087	1.940	-0.556	-2.051	n/a

Table II
Descriptive Statistics on Slant in National and Local News

This table summarizes the distribution of slant by newspaper. *Slant* is the ratio of the number of negative words to the total number of words used in a composite story reported in month *t* multiplied by -100. The sample period is 2002 to 2006. The unit of observation is the firm-month-media outlet. If the distance between the location of a newspaper (see Appendix B) and the location of a firm's headquarters is less than 100 miles, then the firm is considered a local firm for the corresponding newspaper.

	<i>Boston Globe</i>	<i>Chicago Sun Times</i>	<i>Denver Post</i>	<i>Pittsburgh Post- Gazette</i>	<i>Seattle Post- Intelligencer</i>
News on local firms					
Mean	-2.43	-1.14	-1.02	-1.12	-1.30
Median	-1.25	-0.88	-0.74	-0.88	-0.78
SD	2.93	1.09	1.20	0.94	1.58
Number of firms	174	104	68	36	70
Number of composite stories	1,109	834	299	423	879
News on nonlocal firms					
Mean	-2.14	-1.49	-0.90	-0.95	-1.89
Median	-0.95	-1.05	-0.60	-0.67	-0.93
SD	2.81	1.58	1.16	1.01	2.52
Number of firms	602	585	261	560	393
Number of composite stories	2,268	2,504	576	2,198	1,328
	<i>San Francisco Chronicle</i>	<i>St. Louis Post- Dispatch</i>	<i>Washington Post</i>	<i>Dow Jones</i>	<i>Wall Street Journal</i>
News on local firms					
Mean	-0.87	-0.99	-1.21		
Median	-0.6	-0.69	-0.85		
SD	1.00	1.23	1.37		
Number of firms	149	42	143		
Number of composite stories	983	759	1,370		
News on nonlocal firms					
Mean	-1.08	-1.14	-1.05	-1.37	-3.19
Median	-0.65	-0.70	-0.72	-1.05	-1.92
SD	1.39	1.41	1.24	1.24	3.13
Number of firms	226	763	736	5,274	2,780
Number of composite stories	554	2,900	3,252	80,634	22,115

5.2 Table II: descriptive statistics on slant by outlet

- **Read.** This table reports average slant for local and nonlocal firms separately by newspaper.
- Wall Street Journal coverage is the most negative on average, with mean slant around -3.19.
- Across local papers, raw differences between local and nonlocal stories are not always large.
- **Takeaway.** Univariate comparisons are not enough; the main evidence comes from multivariate regressions that hold fixed firm characteristics and event timing.

5.3 Table III: determinants of media slant

- **Read.** This is the core evidence that local papers are more favorable toward local firms.
- With firm-month fixed effects, the coefficient on **Local Firm** is about **0.515**.
- In the richer baseline with controls, the coefficient is about **0.192**.
- With firm fixed effects, it rises to about **0.325**.
- The employee-based catering proxies are negative or unhelpful, and the complexity proxies do not support the constrained-reporting story.

Table III
Determinants of Media Slant, Catering, and Constrained Reporting Hypotheses

This table reports the results of the following pooled OLS regression: $Slant = a + b \times Local\ Firm + c \times Controls + d \times Fixed\ Effects + residual$. The unit of analysis is the firm-month-media outlet. $Slant$ is the ratio of the number of negative words to the total number of words used in a composite story reported in month t multiplied by -100 . The regressors are defined in Appendix C. Heteroskedasticity-robust standard errors are clustered by firm and provided in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Slant	Slant	Slant
Local firm	0.515*** (0.033)	0.192*** (0.055)	0.325*** (0.041)
Log(market value)		0.142*** (0.013)	0.092*** (0.022)
ROA		0.160*** (0.029)	0.168*** (0.039)
Book leverage		-0.035 (0.052)	0.140* (0.077)
Analyst following		0.001 (0.002)	0.002 (0.002)
Inst. ownership (%)		-0.128** (0.050)	0.128** (0.065)
Momentum		0.064*** (0.008)	0.062*** (0.009)
Idiosyncratic volatility		-0.787*** (0.143)	-0.366** (0.146)
Return		0.262*** (0.045)	0.219*** (0.044)
Sales growth		0.006 (0.007)	0.042 (0.063)
Log(employees)		-0.040*** (0.014)	0.126*** (0.038)
Employees/population		-7.603*** (1.999)	-16.886*** (1.657)
Number of segments		0.004 (0.009)	0.003 (0.014)
Plain English		0.016** (0.006)	0.002 (0.007)
Firm fixed effects	Subsumed	No	Included
Media fixed effects	Included	Included	Included
Industry fixed effects	Subsumed	Included	Subsumed
Month fixed effects	Subsumed	Included	Included
Firm-month fixed effects	Included	No	No
Constant	Included	Included	Included
N	124,585	69,334	69,334
Adjusted- R^2	0.43	0.19	0.27

- **Takeaway.** Local firms receive systematically more positive language from local media, and the most plausible alternatives do not explain it well.

5.4 Table IV: media slant and the advertising hypothesis

Table IV
Media Slant and Advertising Hypothesis

The first three columns of the table report the results of the following pooled OLS regression: $Slant = a + b \times Local\ Firm + c \times Log\ (Local\ Advertising) + d \times Log\ (National\ Advertising) + e \times Controls + f \times Fixed\ Effects + residual$. The last three columns of the table report the first- and second-stage results of the 2SLS regression that uses the log of industry local and national advertising spending for *Local Advertising* and *National Advertising* as instruments. We exclude the firm's contribution to industry average when calculating the instruments. The unit of analysis is the firm-month-media outlet. The dependent variable, $Slant$, is the ratio of the number of negative words to the total number of words used in a composite story reported in month t multiplied by -100 . The regressors are defined in Appendix C. Heteroskedasticity-robust standard errors are clustered by firm and provided in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	OLS <i>Slant</i>	OLS <i>Slant</i>	OLS <i>Slant</i>	IV: First Stage <i>Log (Local Advertising)</i>	IV: Second Stage <i>Slant (2SLS)</i>
Local firm	0.325*** (0.071)	0.354*** (0.071)	0.345*** (0.055)	-0.612** (0.259)	-1.559*** (0.453)
Log(market value)	0.152*** (0.023)	0.126*** (0.023)	0.124*** (0.014)	0.601*** (0.156)	1.373*** (0.187)
ROA	0.304 (0.139)	0.242* (0.137)	0.250** (0.104)	-0.885 (1.264)	-1.697 (1.178)
Book leverage	0.024 (0.118)	0.048 (0.112)	0.044 (0.057)	-0.839 (0.660)	-1.063 (1.006)
Analyst following	0.003 (0.002)	0.003 (0.002)	0.003 (0.002)	0.008 (0.016)	0.015 (0.025)
Inst. ownership (%)	-0.307*** (0.091)	-0.256*** (0.088)	-0.254*** (0.057)	-0.352 (0.671)	-3.200*** (0.794)
Momentum	0.091*** (0.023)	0.094*** (0.022)	0.091*** (0.020)	0.059 (0.117)	0.107*** (0.163)
Idiosyncratic volatility	-1.033*** (0.307)	-1.138*** (0.309)	-1.143*** (0.216)	-1.086 (2.162)	7.669*** (2.638)
Return	0.099 (0.093)	0.101 (0.085)	0.099 (0.085)	-0.108 (0.246)	0.098 (0.088)

(continued)

Table IV—Continued

	OLS <i>Slant</i>	OLS <i>Slant</i>	OLS <i>Slant</i>	IV: First Stage <i>Log (National Advertising)</i>	IV: Second Stage <i>Slant (2SLS)</i>
Sales growth	0.487*** (0.156)	0.482*** (0.148)	0.462*** (0.119)	0.321 (1.111)	0.477*** (1.558)
Log (employees)	-0.029 (0.028)	-0.051* (0.028)	-0.061*** (0.014)	1.104*** (0.177)	0.684*** (0.233)
Employees/population	-4.907** (3.187)	-7.469** (3.136)	-6.094** (2.359)	3.208 (14.976)	-8.737*** (23.933)
Number of segments	-0.012 (0.013)	-0.017 (0.013)	-0.016** (0.007)	0.196 (0.079)	0.273** (0.135)
Plain English	0.020** (0.010)	0.016* (0.010)	0.016*** (0.006)	0.130* (0.067)	0.156 (0.095)
Log (national advertising)		0.014*** (0.003)	0.014*** (0.002)		-0.056 (0.054)
Log (local advertising)		0.011*** (0.004)	0.011*** (0.003)		0.057*** (0.021)
IV: Log (industry national advertising)				0.457*** (0.159)	-0.27 (0.245)
IV: Log (industry local advertising)				0.055 (0.057)	0.559*** (0.103)
Media-month fixed effects	No	No	Included	No	No
Media fixed effects	Included	Included	Included	Included	Included
Industry fixed effects	Included	Included	Included	Included	Included
Month fixed effects	Included	Included	Subsumed	Included	Included
Constant	Included	Included	Included	Included	Included
N	28,383	28,383	28,383	28,378	28,378
Adjusted- R^2	0.22	0.22	0.24	0.47	0.52
F -test				41.51	158.64
Cragg-Donald Wald F Statistics					40.25

- **Read.** This table adds local and national newspaper advertising to the slant regression and then instruments for them.
- In OLS, both local and national advertising are positively associated with slant.
- In IV, the coefficient on **local advertising** is about **0.057** and statistically significant.

Table V
Propensity Score Matching Estimates: Local Advertising
Panel A reports the results of the propensity score matching estimates (Column 1) and the sample means of the treatment (Column 2) and control (Column 3) samples for the corresponding variables. The last column reports the p -values of the difference in means. Heteroskedasticity-robust standard errors are clustered by firm. Panel B reports the average treatment effects on *Slant*, where the treatment is defined as "Local Advertising > 0." *Slant* is the ratio of the number of negative words to the total number of words used in a composite story reported in month t multiplied by -100 . Matching estimates use the Gaussian kernel with a fixed bandwidth of 0.10. Standard errors (in parentheses) are obtained using 500 bootstrap repetitions. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Propensity Score Model Estimates for Local Media Advertising Spending				
	Dep. Variable = 1 if Local Advertising > 0	Treatment Sample Mean	Control Sample Mean	Pr (difference)
Log (market value)	0.286***	7.480	8.981	0.000
ROA	-0.559	0.037	0.052	0.487
Book leverage	-0.340	0.461	0.461	0.997
Analyst following	0.001	7.221	11.294	0.000
Inst. ownership (%)	-0.577***	0.663	0.639	0.276
Momentum	-0.082*	0.215	0.149	0.269
Idiosyncratic volatility	2.132***	0.119	0.109	0.414
Return	-0.043	0.012	0.005	0.319
Sales growth	-0.231	1.056	1.059	0.766
Local firm	-0.412***	0.059	0.066	0.671
Log (national advertising)	0.090***	7.982	11.995	0.000
Log (employees)	0.014	8.559	9.680	0.000
Employees/population	7.083	0.000	0.003	0.043
Number of segments	0.065***	2.875	3.248	0.317
Plain English	0.018	0.495	0.636	0.753
Media fixed effects	Included			
Industry fixed effects	Included			
Month fixed effects	Included			
Constant	Included			
Number of treated observations	14,918			
Number of untreated observations	13,460			
Panel B: Average Treatment Effect (ATE) on <i>Slant</i> for Local Media Advertising Spending				
				ATE
Local media advertising spending (expenditure > 0 vs. = 0)				13.2%*** (0.042)

- The coefficient on **national advertising** becomes about -0.056 and statistically insignificant.
- **Takeaway.** Once endogeneity is addressed, the evidence points specifically to **local advertising** as the causal driver of positive local-media tone.

5.5 Table V: propensity score matching

- **Read.** Firms with positive local advertising are matched to similar firms without local advertising.
- The average treatment effect of local advertising on slant is about **13.2%**.
- **Takeaway.** The advertising-slant relation is not just an artifact of obvious observable differences across firms.

5.6 Table VI: Craigslist quasi-natural experiment

- **Read.** After Craigslist enters Pittsburgh and St. Louis, the interaction between local advertising and the post-entry indicator becomes positive.
- For the all-article sample, the coefficient on $\log(\text{Local Advertising}) \times \text{Craigslist}$ is about **0.020**.
- For articles about Pittsburgh and St. Louis firms only, it rises to about **0.045**.
- **Takeaway.** When local newspapers face more pressure on ad revenues, they appear to supply more slant in response to local advertising dollars.

Table VI
Slant and the Effect of Competition from Craigslist

This table reports the results of the following pooled OLS regression: $Slant = a + b \times Craigslist + Log(National\ Advertising) + c \times Log(Local\ Advertising) + d \times Log(National\ Advertising) \times Craigslist + e \times Log(Local\ Advertising) \times Craigslist + f \times Controls + g \times Fixed\ Effects + residual$. The unit of analysis is the firm-month-media outlet. *Slant* is the ratio of the number of negative words to the total number of words used in a composite story reported by the *Pittsburgh Gazette* and the *St. Louis Post-Dispatch* in month t (multiplied by -100). *Craigslist* is an indicator variable that takes a value of one after October 2003. Heteroskedasticity-robust standard errors are clustered by firm and provided in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Slant: All Articles</i>	<i>Slant: Articles about Pittsburgh and St. Louis Firms Only</i>
Craigslist	-0.013 (0.201)	0.380*** (0.129)
Log (national advertising)	0.008 (0.017)	0.021 (0.019)
Log (local advertising)	-0.007 (0.011)	-0.018 (0.013)
Log (national advertising) $\times$ Craigslist	-0.005 (0.016)	-0.045*** (0.012)
Log (local advertising) $\times$ Craigslist	0.020* (0.011)	0.045*** (0.016)
Local firm	0.391*** (0.149)	-0.205 (0.129)
Log (market value)	0.099*** (0.042)	-0.362 (0.252)
ROA	0.063 (0.386)	-0.141 (0.646)
Book leverage	-0.324 (0.252)	-1.209 (0.815)
Analyst following	0.002 (0.005)	-0.014 (0.033)
Inst. ownership (%)	0.149 (0.236)	0.629 (0.547)
Momentum	0.124* (0.064)	0.045 (0.157)
Idiosyncratic volatility	-1.494* (0.820)	1.161 (0.998)
Return	0.792*** (0.368)	0.451 (0.345)
Sales growth	0.653 (0.615)	4.655*** (1.201)
Log (employees)	-0.049 (0.074)	0.943*** (0.340)
Employees/population	0.011 (5.081)	-81.576*** (11.537)
Number of segments	0.004 (0.021)	0.056 (0.057)
Plain English	-0.027 (0.021)	0.130*** (0.045)

(continued)

Table VI—Continued

	<i>Slant: All Articles</i>	<i>Slant: Articles about Pittsburgh and St. Louis Firms Only</i>
Media fixed effects	Included	Included
Month fixed effects	Included	Included
Industry fixed effects	Included	Included
Constant	Included	Included
N	1,848	215
Adjusted- R^2	0.10	0.34

5.7 Table VII: local media-slant trading profits

- **Read.** A long-short portfolio that buys firms with **low abnormal local slant** and shorts firms with **high abnormal local slant** earns significant alpha.
- In the six-factor model including **NOCOV**, the alpha is about **0.46% per month**, or roughly **5.52% per year**.
- **Takeaway.** Stocks receiving unusually favorable local hype seem overpriced and subsequently underperform.

5.8 Table VIII: newspaper slant and firm value

- **Read.** This is the main Tobin's Q regression.
- The coefficient on annual local slant is about **0.024–0.025**.
- The coefficient on annual national slant is about **0.060–0.064**.
- A one-standard-deviation increase in annual local slant is associated with about **4.29%** higher firm value.
- **Takeaway.** Even after controlling for national tone and the quantity of local and national coverage, local slant still relates positively to firm valuation.

Table VIII
Newspaper Slant and Firm Value

This table reports the results of the following pooled OLS regression: $\text{Log of Tobin's } Q = a + b \times \text{Annual Local Slant} + c \times \text{Annual National Slant} + d \times \text{Annual Local Coverage} + e \times \text{Annual National Coverage} + f \times \text{Controls} + g \times \text{Fixed Effects} + \text{residual}$. The unit of analysis is the firm-year. The regressors are defined in Appendix C. Heteroskedasticity-robust standard errors are clustered by firm and provided in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Ln(Q)	Ln(Q)
Annual local slant	0.024*** (0.009)	0.025*** (0.006)
Annual national slant	0.064*** (0.014)	0.060*** (0.011)
Annual local coverage	0.006 (0.014)	0.009 (0.012)
Annual nat'l coverage	0.127*** (0.020)	0.103*** (0.014)
Log (total assets)	-0.158*** (0.016)	-0.132*** (0.010)
Return on assets	0.093 (0.100)	0.061*** (0.027)
CAPX-to-sales	-0.724*** (0.126)	-0.633*** (0.089)
Book leverage	0.262*** (0.085)	0.172*** (0.039)
R&D-to-sales	0.318*** (0.144)	0.074 (0.105)
Advertising-to-sales	0.72 (0.515)	0.298 (0.266)
Analyst following	0.023*** (0.002)	0.021*** (0.002)
Inst. ownership (%)	0.359*** (0.060)	0.368*** (0.044)
Momentum	0.117*** (0.019)	0.112*** (0.011)
Idiosyncratic volatility	-1.067*** (0.197)	-0.852*** (0.143)
Sales growth	0.643*** (0.109)	0.563*** (0.077)
Industry fixed effects	Included	No
Year fixed effects	Included	Included
Constant	Included	Included
Four-digit industry fixed effects	No	Included
N	3,020	3,020
Adjusted-R ²	0.54	0.65

Table IX
Newspaper Slant, Coverage, and Firm Value: Subsample Analysis

This table reports results of the following pooled OLS regression for the subsamples described in the column headings: $\text{Log of Tobin's } Q = a + b \times \text{Annual Local Slant} + c \times \text{Annual National Slant} + d \times \text{Annual Local Coverage} + e \times \text{Annual National Coverage} + f \times \text{Controls} + g \times \text{Fixed Effects} + \text{residual}$. The unit of analysis is the firm-year. The regressors are defined in Appendix C. Heteroskedasticity-robust standard errors are clustered by firm and provided in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	Institutional Ownership		Size		Analyst Following		Forecast Dispersion		Idiosyncratic Volatility		Liquidity	
	Low Ln(Q)	High Ln(Q)	Small Ln(Q)	Large Ln(Q)	Low Ln(Q)	High Ln(Q)	Low Ln(Q)	High Ln(Q)	Low Ln(Q)	High Ln(Q)	Low Ln(Q)	High Ln(Q)
Annual local slant	0.026*** (0.010)	0.012 (0.008)	0.018* (0.011)	0.01 (0.010)	0.027*** (0.011)	0.005 (0.010)	0.015 (0.012)	0.024*** (0.011)	0.004 (0.010)	0.023*** (0.011)	0.028*** (0.008)	0.002 (0.011)
Annual nat'l slant	0.053*** (0.020)	0.071*** (0.016)	0.051*** (0.017)	0.083*** (0.020)	0.055*** (0.017)	0.071*** (0.021)	0.070*** (0.021)	0.056*** (0.016)	0.076*** (0.018)	0.051*** (0.018)	0.072*** (0.021)	0.032*** (0.015)
Annual local coverage	0.01 (0.020)	-0.002 (0.016)	-0.009 (0.021)	0.02 (0.016)	0.006 (0.018)	0.016 (0.019)	0.013 (0.017)	0.001 (0.016)	0.011 (0.016)	-0.006 (0.021)	-0.023 (0.022)	0.006 (0.014)
Annual nat'l coverage	0.159*** (0.028)	0.064*** (0.024)	0.083*** (0.024)	0.133*** (0.030)	0.137*** (0.024)	0.098*** (0.031)	0.091*** (0.031)	0.128*** (0.022)	0.117*** (0.028)	0.085*** (0.024)	0.164*** (0.027)	0.060*** (0.024)
Control variables used in Table VIII	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
Adjusted-R ²	0.56	0.59	0.54	0.74	0.49	0.69	0.72	0.51	0.72	0.50	0.46	0.76

5.9 Table IX: subsample analysis

- **Read.** This table shows where local slant matters most for value.
- The effect is strong for:
 - firms with **low institutional ownership**,
 - **small** firms,
 - firms with **low analyst following**,
 - firms with **high analyst forecast dispersion**,
 - firms with **high idiosyncratic volatility**,
 - firms with **low liquidity**.
- **Takeaway.** The pricing effect is concentrated exactly where local investors are more likely to matter and arbitrage is harder.

6 Short summary

- **Method.** Build a text-based measure of tone from newspaper stories, relate it to geography and advertising, and then connect it to prices and valuations.
- **Identification.** Use within-firm cross-outlet comparisons, IV based on industry-year advertising averages, propensity-score matching, and a Craigslist quasi-experiment.
- **Main result.** Local newspapers write less negatively about nearby firms, and that local cheerleading is strongly related to local advertising expenditures.
- **Valuation result.** Abnormally positive local slant predicts lower subsequent returns and higher contemporaneous Tobin's Q, especially for firms with poor information environments or high arbitrage frictions.

7 Conclusion

7.1 Core conclusion

- The paper shows that local newspapers do not behave like neutral information processors. They write systematically more positive stories about nearby firms.
- The most convincing explanation in the data is **local advertising pressure**, not employee catering and not simple reporting constraints.
- This slant is not just statistically detectable; it appears to matter for both **stock returns** and **firm value**.

7.2 Economic intuition

- Local newspapers depend heavily on advertising revenue.
- Local firms can therefore matter financially to local newspapers in a way that nonlocal firms or national advertisers do not.
- When a local paper softens negative language about local advertisers, local investors may receive a systematically more favorable signal about nearby firms.
- If those investors are marginal investors and arbitrage is limited, prices can be pushed away from fundamentals.

7.3 Takeaway

This paper is a clean example of how **information intermediaries have incentives**. In finance, the relevant question is not only whether investors receive news, but also **who produces the news, what conflicts they face, and how those conflicts shape tone**. Gurun and Butler's main lesson is that local media slant is economically meaningful and that part of home bias may come from investors seeing *more favorable*, not necessarily more accurate, information about local firms.